# A Database-Driven Chit Fund Management and Auction System

## GROUP-10

MEMBER 1:

NAME : Shasank SK

SRN: PES2UG24CS460

MEMBER 2:

NAME: Samartha MS

SRN: PES2UG24CS435

# Already Executed DDL Statements

```
mysql> prompt PES2UG24CS460_PES2UG24CS435_G10$
PROMPT set to 'PES2UG24CS460_PES2UG24CS435_G10$'
PES2UG24CS460_PES2UG24CS435_G10$SHOW TABLES;
+----------------------+
| Tables_in_chitvault |
+----------------------+
| bids                 |
| chitgroups           |
| defaulters           |
| groupmembers         |
| kycdocuments         |
| monthlycycles        |
| notifications        |
| payments             |
| payouts              |
| users                |
+----------------------+
10 rows in set (0.00 sec)
```



```
PES2UG24CS460_PES2UG24CS435_G10$SHOW TRIGGERS;
+-----------------------+--------+--------+-----------------------
```

```
PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE Users ADD COLUMN city VARCHAR(50) AFTER phone;
Query OK, 0 rows affected (0.29 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE Users ADD CONSTRAINT chk_phone_len CHECK (CHAR_LENGTH(phone) >= 10);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE GroupMembers ADD CONSTRAINT uq_group_user UNIQUE (group_id, user_id);
Query OK, 0 rows affected, 1 warning (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 1

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE Payments ADD CONSTRAINT chk_payment_amt CHECK (amount_paid >= 0 AND penalty_amount >= 0);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE Bids ADD CONSTRAINT chk_bid_amt CHECK (bid_amount > 0);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE ChitGroups ADD CONSTRAINT chk_min_members CHECK (max_members > 1);
Query OK, 0 rows affected (0.13 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE Notifications MODIFY type VARCHAR(30) NOT NULL DEFAULT 'general';
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE TABLE AuditLog (
    ->     log_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     table_name  VARCHAR(50)  NOT NULL,
    ->     record_id   INT          NOT NULL,
    ->     action      VARCHAR(30)  NOT NULL,
    ->     old_value   VARCHAR(100),
    ->     new_value   VARCHAR(100),
    ->     changed_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE TABLE TempScratch (id INT PRIMARY KEY, note VARCHAR(50));
Query OK, 0 rows affected (0.06 sec)

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE TempScratch ADD COLUMN created_on DATE;
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE TempScratch RENAME COLUMN note TO remarks;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$ALTER TABLE TempScratch DROP COLUMN remarks;
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$DROP TABLE TempScratch;
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE INDEX idx_payments_status_cycle ON Payments (status, cycle_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$CREATE INDEX idx_bids_cycle_amount     ON Bids (cycle_id, bid_amount DESC);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$CREATE INDEX idx_cycles_group_status   ON MonthlyCycles (group_id, status);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$CREATE INDEX idx_defaulters_user_stat  ON Defaulters (user_id, resolved_status);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$CREATE INDEX idx_users_name            ON Users (name);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$CREATE FULLTEXT INDEX ft_notif_message ON Notifications (message);
Query OK, 0 rows affected, 1 warning (0.33 sec)
Records: 0  Duplicates: 0  Warnings: 1

PES2UG24CS460_PES2UG24CS435_G10$DROP INDEX idx_users_name ON Users;        -- DROP INDEX demo
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$SHOW INDEX FROM Payments;
+----------+------------+-------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name                      | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+----------+------------+-------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| payments |          0 | PRIMARY                       |            1 | payment_id  | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| payments |          0 | cycle_id                      |            1 | cycle_id    | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| payments |          0 | cycle_id                      |            2 | user_id     | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| payments |          1 | user_id                       |            1 | user_id     | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| payments |          1 | idx_payments_status_cycle     |            1 | status      | A         |           0 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| payments |          1 | idx_payments_status_cycle     |            2 | cycle_id    | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+----------+------------+-------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
6 rows in set (0.02 sec)
```

# DML

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Users (user_id, name, email, phone, city, password_hash, role, kyc_verified) VALUES
    -> (1,  'Admin Rao',     'admin@chitvault.com',   '9000000001', 'Bengaluru', 'hash_admin', 'admin',     TRUE),
    -> (2,  'Ramesh Kumar',  'ramesh@chitvault.com',  '9000000002', 'Bengaluru', 'hash_r',     'organizer', TRUE),
    -> (3,  'Lakshmi Devi',  'lakshmi@chitvault.com', '9000000003', 'Mysuru',    'hash_l',     'organizer', TRUE),
    -> (4,  'Arjun Nair',    'arjun@mail.com',        '9000000004', 'Bengaluru', 'hash_a',     'member',    FALSE),
    -> (5,  'Priya Sharma',  'priya@mail.com',        '9000000005', 'Chennai',   'hash_p',     'member',    FALSE),
    -> (6,  'Karthik Reddy', 'karthik@mail.com',      '9000000006', 'Hyderabad', 'hash_k',     'member',    FALSE),
    -> (7,  'Sneha Iyer',    'sneha@mail.com',        '9000000007', 'Bengaluru', 'hash_s',     'member',    FALSE),
    -> (8,  'Rahul Verma',   'rahul@mail.com',        '9000000008', 'Pune',      'hash_rv',    'member',    FALSE),
    -> (9,  'Divya Menon',   'divya@mail.com',        '9000000009', 'Kochi',     'hash_d',     'member',    FALSE),
    -> (10, 'Mohan Das',     'mohan@mail.com',        '9000000010', 'Mysuru',    'hash_m',     'member',    FALSE),
    -> (11, 'Test User',     'test@mail.com',         '9000000011', 'Delhi',     'hash_t',     'member',    FALSE);
Query OK, 11 rows affected (0.01 sec)
Records: 11  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO KYCDocuments (user_id, doc_type, doc_number, verified_status) VALUES
    -> (4,  'Aadhar', '1111-2222-3333', 'verified'),
    -> (5,  'PAN',    'ABCDE1234F',     'verified'),
    -> (6,  'Aadhar', '2222-3333-4444', 'verified'),
    -> (7,  'PAN',    'BCDEF2345G',     'verified'),
    -> (8,  'Aadhar', '3333-4444-5555', 'verified'),
    -> (9,  'PAN',    'CDEFG3456H',     'verified'),
    -> (10, 'Aadhar', '4444-5555-6666', 'verified'),
    -> (11, 'Aadhar', '5555-6666-7777', 'pending');
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO ChitGroups (group_id, group_name, organizer_id, total_amount, monthly_installment,
    ->                         duration_months, max_members, start_date, status) VALUES
    -> (1, 'Gold Savers Circle',   2, 100000, 12500, 8,  8,  '2026-01-01', 'active'),
    -> (2, 'Silver Monthly Club',  3,  30000,  5000, 6,  6,  '2026-02-01', 'active'),
    -> (3, 'Platinum Future Fund', 2, 240000, 20000, 12, 12, '2026-12-01', 'upcoming');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO GroupMembers (group_id, user_id, joined_date) VALUES
    -> (1,3,'2025-12-20'),(1,4,'2025-12-21'),(1,5,'2025-12-22'),(1,6,'2025-12-22'),
    -> (1,7,'2025-12-23'),(1,8,'2025-12-24'),(1,9,'2025-12-25'),(1,10,'2025-12-26'),
    -> (2,4,'2026-01-15'),(2,5,'2026-01-16'),(2,6,'2026-01-17'),
    -> (2,7,'2026-01-18'),(2,8,'2026-01-19'),(2,9,'2026-01-20');
Query OK, 14 rows affected (0.01 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO MonthlyCycles (cycle_id, group_id, month_number, cycle_date, status) VALUES
    -> (1, 1, 1, '2026-01-05', 'bidding'),
    -> (2, 1, 2, '2026-02-05', 'bidding'),
    -> (3, 1, 3, '2026-03-05', 'bidding'),
    -> (4, 2, 1, '2026-02-10', 'bidding'),
    -> (5, 2, 2, '2026-03-10', 'collection');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    -> SELECT 1, user_id, 12500, '2026-01-04', 'paid' FROM GroupMembers WHERE group_id = 1;
Query OK, 8 rows affected (0.02 sec)
Records: 8  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    -> SELECT 2, user_id, 12500, '2026-02-04', 'paid' FROM GroupMembers WHERE group_id = 1;
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    -> SELECT 3, user_id, 12500, '2026-03-04', 'paid' FROM GroupMembers WHERE group_id = 1 AND user_id <> 10;
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status, penalty_amount)
    -> VALUES (3, 10, 0, '2026-03-06', 'late', 0);
Query OK, 1 row affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    -> SELECT 4, user_id, 5000, '2026-02-09', 'paid' FROM GroupMembers WHERE group_id = 2;
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    -> SELECT 5, user_id, 5000, '2026-03-09', 'paid' FROM GroupMembers WHERE group_id = 2 AND user_id IN (4,5,6);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Bids (cycle_id, user_id, bid_amount, bid_time) VALUES
    -> (1, 4, 8000, '2026-01-05 10:00:00'), (1, 5, 9500, '2026-01-05 10:05:00'),
    -> (1, 6, 7000, '2026-01-05 10:10:00'), (1, 7, 9000, '2026-01-05 10:15:00'),
    -> (2, 6, 8500, '2026-02-05 10:00:00'), (2, 7, 9200, '2026-02-05 10:07:00'),
    -> (2, 8, 7500, '2026-02-05 10:12:00'), (2, 3, 6000, '2026-02-05 10:20:00'),
    -> (3, 3, 9000, '2026-03-05 10:00:00'), (3, 4, 8800, '2026-03-05 10:10:00'),
    -> (3, 6, 9100, '2026-03-05 10:15:00'),
    -> (4, 4, 3000, '2026-02-10 11:00:00'), (4, 5, 3500, '2026-02-10 11:05:00'),
    -> (4, 9, 2800, '2026-02-10 11:10:00');
Query OK, 14 rows affected (0.01 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Notifications (user_id, message, type, read_status) VALUES
    -> (4, 'Welcome to ChitVault. Your KYC is verified.',           'welcome',  TRUE),
    -> (5, 'Auction for cycle 1 opens tomorrow at 10 AM.',          'auction',  FALSE),
    -> (6, 'Payment reminder: monthly installment is due.',         'reminder', FALSE),
    -> (7, 'Auction results will be announced after bidding ends.', 'auction',  TRUE),
    -> (8, 'Payment reminder: please pay before the cycle date.',   'reminder', FALSE);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
PES2UG24CS460_PES2UG24CS435_G10$UPDATE Users
    -> SET kyc_verified = TRUE
    -> WHERE user_id IN (SELECT user_id FROM KYCDocuments WHERE verified_status = 'verified');
Query OK, 7 rows affected (0.02 sec)
Rows matched: 7  Changed: 7  Warnings: 0
```

# STORED FUNCTIONS

```
PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_member_count//
Query OK, 0 rows affected, 1 warning (0.02 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE FUNCTION fn_member_count(p_group INT) RETURNS INT
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE v INT;
    ->     SELECT COUNT(*) INTO v FROM GroupMembers
    ->     WHERE group_id = p_group AND slot_status <> 'exited';
    ->     RETURN v;
    -> END//
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_calc_penalty//
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_calc_penalty//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE FUNCTION fn_calc_penalty(p_installment DECIMAL(10,2), p_days_late INT) RETURNS DECIMAL(1
0,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     -- 2% flat penalty + Rs.10 per day late
    ->     RETURN ROUND(p_installment * 0.02 + GREATEST(p_days_late, 0) * 10, 2);
    -> END//
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_total_paid//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE FUNCTION fn_total_paid(p_user INT, p_group INT) RETURNS DECIMAL(12,2)
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE v DECIMAL(12,2);
    ->     SELECT IFNULL(SUM(p.amount_paid), 0) INTO v
    ->     FROM Payments p JOIN MonthlyCycles mc ON mc.cycle_id = p.cycle_id
    ->     WHERE p.user_id = p_user AND mc.group_id = p_group;
    ->     RETURN v;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_dividend//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE FUNCTION fn_dividend(p_group INT, p_bid DECIMAL(10,2)) RETURNS DECIMAL(10,2)
    -> READS SQL DATA
    -> BEGIN
    ->     -- winning bid (discount) is shared equally among all members
    ->     RETURN ROUND(p_bid / fn_member_count(p_group), 2);
    -> END//
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$DROP FUNCTION IF EXISTS fn_risk_level//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE FUNCTION fn_risk_level(p_user INT) RETURNS VARCHAR(10)
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE c INT;
    ->     SELECT COUNT(*) INTO c FROM Defaulters WHERE user_id = p_user;
    ->     RETURN CASE WHEN c = 0 THEN 'LOW' WHEN c = 1 THEN 'MEDIUM' ELSE 'HIGH' END;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

# TRIGGERS (3 already exist)

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_limit_members//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_limit_members
    -> BEFORE INSERT ON GroupMembers
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE mx INT;
    ->     SELECT max_members INTO mx FROM ChitGroups WHERE group_id = NEW.group_id;
    ->     IF fn_member_count(NEW.group_id) >= mx THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Group is full: max_members reached';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_validate_bid//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_validate_bid
    -> BEFORE INSERT ON Bids
    -> FOR EACH ROW
    -> FOLLOWS trg_prevent_repeat_bid
    -> BEGIN
    ->     DECLARE v_status VARCHAR(20);
    ->     DECLARE v_group  INT;
    ->     DECLARE v_total  DECIMAL(12,2);
    ->     DECLARE v_active INT;
    ->
    ->     SELECT mc.status, mc.group_id, cg.total_amount INTO v_status, v_group, v_total
    ->     FROM MonthlyCycles mc JOIN ChitGroups cg ON cg.group_id = mc.group_id
    ->     WHERE mc.cycle_id = NEW.cycle_id;
    ->
    ->     IF v_status <> 'bidding' THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Bidding is not open for this cycle';
    ->     END IF;
    ->     IF NEW.bid_amount > v_total * 0.30 THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Bid exceeds 30% of the chit value';
    ->     END IF;
    ->     SELECT COUNT(*) INTO v_active FROM GroupMembers
    ->     WHERE group_id = v_group AND user_id = NEW.user_id AND slot_status = 'active';
    ->     IF v_active = 0 THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Only active group members can bid';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_auto_penalty//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_auto_penalty
    -> BEFORE INSERT ON Payments
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE v_inst DECIMAL(10,2);
    ->     DECLARE v_due  DATE;
    ->     IF NEW.status = 'late' AND IFNULL(NEW.penalty_amount, 0) = 0 THEN
    ->         SELECT cg.monthly_installment, mc.cycle_date INTO v_inst, v_due
    ->         FROM MonthlyCycles mc JOIN ChitGroups cg ON cg.group_id = mc.group_id
    ->         WHERE mc.cycle_id = NEW.cycle_id;
    ->         SET NEW.penalty_amount =
    ->             fn_calc_penalty(v_inst, DATEDIFF(IFNULL(NEW.payment_date, CURDATE()), v_due));
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_notify_defaulter//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_notify_defaulter
    -> AFTER INSERT ON Defaulters
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE v_pending INT;
    ->     INSERT INTO Notifications (user_id, message, type)
    ->     VALUES (NEW.user_id,
    ->             CONCAT('You have been flagged as a defaulter for cycle ', NEW.cycle_id,
    ->                    '. Please clear your dues to avoid exclusion.'), 'default');
    ->
    ->     SELECT COUNT(*) INTO v_pending FROM Defaulters
    ->     WHERE user_id = NEW.user_id AND group_id = NEW.group_id AND resolved_status = 'pending';
    ->     IF v_pending >= 2 THEN
    ->         UPDATE GroupMembers SET slot_status = 'defaulted'
    ->         WHERE user_id = NEW.user_id AND group_id = NEW.group_id;
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_notify_payout//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_notify_payout
    -> AFTER INSERT ON Payouts
    -> FOR EACH ROW
    -> FOLLOWS trg_lock_winner
    -> BEGIN
    ->     INSERT INTO Notifications (user_id, message, type)
    ->     VALUES (NEW.winner_id,
    ->             CONCAT('Congratulations! You won the auction for cycle ', NEW.cycle_id,
    ->                    '. Payout amount: Rs. ', NEW.amount_disbursed), 'payout');
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP TRIGGER IF EXISTS trg_audit_group_status//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE TRIGGER trg_audit_group_status
    -> AFTER UPDATE ON ChitGroups
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.status <> NEW.status THEN
    ->         INSERT INTO AuditLog (table_name, record_id, action, old_value, new_value)
    ->         VALUES ('ChitGroups', NEW.group_id, 'STATUS_CHANGE', OLD.status, NEW.status);
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.02 sec)
```

# PROCEDURES

```
PES2UG24CS460_PES2UG24CS435_G10$DROP PROCEDURE IF EXISTS sp_add_member//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_add_member(IN p_group INT, IN p_user INT)
    -> BEGIN
    ->     DECLARE v_kyc BOOLEAN;
    ->     SELECT kyc_verified INTO v_kyc FROM Users WHERE user_id = p_user;
    ->     IF v_kyc IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'User does not exist';
    ->     END IF;
    ->     IF v_kyc = FALSE THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'KYC not verified: cannot join group';
    ->     END IF;
    ->     INSERT INTO GroupMembers (group_id, user_id) VALUES (p_group, p_user);
    ->     SELECT CONCAT('User ', p_user, ' added to group ', p_group) AS result;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP PROCEDURE IF EXISTS sp_record_payment//
Query OK, 0 rows affected, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_record_payment(IN p_cycle INT, IN p_user INT, IN p_amount DECIMAL(10,2), IN p_date DATE)
    -> BEGIN
    ->     DECLARE v_inst   DECIMAL(10,2);
    ->     DECLARE v_due    DATE;
    ->     DECLARE v_status VARCHAR(10);
    ->
    ->     SELECT cg.monthly_installment, mc.cycle_date INTO v_inst, v_due
    ->     FROM MonthlyCycles mc JOIN ChitGroups cg ON cg.group_id = mc.group_id
    ->     WHERE mc.cycle_id = p_cycle;
    ->
    ->     IF v_inst IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid cycle';
    ->     END IF;
    ->     IF p_amount < v_inst THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Amount is less than the monthly installment';
    ->     END IF;
    ->
    ->     SET v_status = IF(p_date > v_due, 'late', 'paid');
    ->     INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    ->     VALUES (p_cycle, p_user, p_amount, p_date, v_status);   -- trg_auto_penalty fills penalty if late
    ->
    ->     SELECT CONCAT('Payment recorded as ', v_status) AS result;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_place_bid(IN p_cycle INT, IN p_user INT, IN p_amount DECIMAL(10,2))
    -> BEGIN
    ->     INSERT INTO Bids (cycle_id, user_id, bid_amount) VALUES (p_cycle, p_user, p_amount);
    ->     SELECT 'Bid placed successfully' AS result;
    -> END//
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$DROP PROCEDURE IF EXISTS sp_close_cycle//
Query OK, 0 rows affected, 1 warning (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_close_cycle(IN p_cycle INT)
    -> BEGIN
    ->     DECLARE v_group  INT;
    ->     DECLARE v_status VARCHAR(20);
    ->     DECLARE v_total  DECIMAL(12,2);
    ->     DECLARE v_bid_id INT;
    ->     DECLARE v_winner INT;
    ->     DECLARE v_bid    DECIMAL(10,2);
    ->     DECLARE v_pool   DECIMAL(12,2);
    ->
    ->     DECLARE EXIT HANDLER FOR SQLEXCEPTION
    ->     BEGIN
    ->         ROLLBACK;
    ->         RESIGNAL;
    ->     END;
    ->
    ->     SELECT mc.group_id, mc.status, cg.total_amount INTO v_group, v_status, v_total
    ->     FROM MonthlyCycles mc JOIN ChitGroups cg ON cg.group_id = mc.group_id
    ->     WHERE mc.cycle_id = p_cycle;
    ->
    ->     IF v_group IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid cycle';
    ->     END IF;
    ->     IF v_status = 'closed' THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Cycle is already closed';
    ->     END IF;
    ->
    ->     START TRANSACTION;
    ->
    ->     -- highest bid wins; tie -> earliest bid
    ->     SELECT bid_id, user_id, bid_amount INTO v_bid_id, v_winner, v_bid
    ->     FROM Bids WHERE cycle_id = p_cycle
    ->     ORDER BY bid_amount DESC, bid_time ASC, bid_id ASC LIMIT 1;
    ->
    ->     IF v_bid_id IS NULL THEN
    ->         SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'No bids placed for this cycle';
    ->     END IF;
    ->
    ->     UPDATE Bids SET is_winner = TRUE WHERE bid_id = v_bid_id;
    ->
    ->     SELECT IFNULL(SUM(amount_paid), 0) INTO v_pool FROM Payments WHERE cycle_id = p_cycle;
    ->
    ->     INSERT INTO Payouts (cycle_id, winner_id, amount_disbursed, dividend_per_member)
    ->     VALUES (p_cycle, v_winner, v_total - v_bid, fn_dividend(v_group, v_bid));
    ->     -- fires trg_lock_winner (sets has_won_payout) and trg_notify_payout
    ->
    ->     UPDATE MonthlyCycles
    ->     SET status = 'closed', winning_bid_amount = v_bid, total_pool_amount = v_pool
    ->     WHERE cycle_id = p_cycle;
    ->
    ->     COMMIT;
    ->
    ->     SELECT p_cycle AS cycle_id, v_winner AS winner_id, v_bid AS winning_bid,
    ->            v_total - v_bid AS amount_disbursed, fn_dividend(v_group, v_bid) AS dividend_per_member;
    -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_mark_defaulters(IN p_cycle INT)
    -> BEGIN
    ->     DECLARE done    INT DEFAULT 0;
    ->     DECLARE v_user  INT;
    ->     DECLARE v_count INT DEFAULT 0;
    ->     DECLARE v_cdate DATE;
    ->     DECLARE cur CURSOR FOR
    ->         SELECT gm.user_id
    ->         FROM GroupMembers gm JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    ->         WHERE mc.cycle_id = p_cycle AND gm.slot_status = 'active'
    ->           AND gm.user_id NOT IN (SELECT user_id FROM Payments WHERE cycle_id = p_cycle);
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     SELECT cycle_date INTO v_cdate FROM MonthlyCycles WHERE cycle_id = p_cycle;
    ->
    ->     OPEN cur;
    ->     read_loop: LOOP
    ->         FETCH cur INTO v_user;
    ->         IF done = 1 THEN LEAVE read_loop; END IF;
    ->
    ->         INSERT INTO Payments (cycle_id, user_id, amount_paid, payment_date, status)
    ->         VALUES (p_cycle, v_user, 0, v_cdate, 'late');           -- penalty auto-calculated
    ->         UPDATE Payments SET status = 'defaulted'
    ->         WHERE cycle_id = p_cycle AND user_id = v_user;           -- fires trg_flag_defaulter
    ->         SET v_count = v_count + 1;
    ->     END LOOP;
    ->     CLOSE cur;
    ->
    ->     -- trg_flag_defaulter stores amount_paid (0) as amount_due; correct it to the installment
    ->     UPDATE Defaulters d JOIN ChitGroups g ON g.group_id = d.group_id
    ->     SET d.amount_due = g.monthly_installment
    ->     WHERE d.amount_due = 0;
    ->
    ->     SELECT v_count AS members_marked_defaulter;
    -> END//
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DROP PROCEDURE IF EXISTS sp_group_summary//
Query OK, 0 rows affected, 1 warning (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$CREATE PROCEDURE sp_group_summary(IN p_group INT)
    -> BEGIN
    ->     SELECT * FROM vw_group_overview WHERE group_id = p_group;
    ->     SELECT u.name, gm.slot_status, gm.has_won_payout
    ->     FROM GroupMembers gm JOIN Users u ON u.user_id = gm.user_id
    ->     WHERE gm.group_id = p_group;
    ->     SELECT month_number, cycle_date, status, winning_bid_amount
    ->     FROM MonthlyCycles WHERE group_id = p_group ORDER BY month_number;
    -> END//
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$DELIMITER ;
```

# VIEWS

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE OR REPLACE VIEW vw_group_overview AS
    -> SELECT cg.group_id, cg.group_name, u.name AS organizer, cg.total_amount,
    ->        cg.monthly_installment, cg.duration_months, cg.max_members,
    ->        fn_member_count(cg.group_id) AS current_members,
    ->        (SELECT COUNT(*) FROM MonthlyCycles mc
    ->         WHERE mc.group_id = cg.group_id AND mc.status = 'closed') AS cycles_completed,
    ->        cg.status
    -> FROM ChitGroups cg JOIN Users u ON u.user_id = cg.organizer_id;
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE OR REPLACE VIEW vw_member_payment_summary AS
    -> SELECT u.user_id, u.name, cg.group_id, cg.group_name,
    ->        COUNT(p.payment_id)               AS payments_made,
    ->        IFNULL(SUM(p.amount_paid), 0)     AS total_paid,
    ->        IFNULL(SUM(p.penalty_amount), 0)  AS total_penalty,
    ->        IFNULL(SUM(p.status = 'late'), 0)     AS late_count,
    ->        IFNULL(SUM(p.status = 'defaulted'), 0) AS default_count
    -> FROM GroupMembers gm
    -> JOIN Users u        ON u.user_id  = gm.user_id
    -> JOIN ChitGroups cg  ON cg.group_id = gm.group_id
    -> LEFT JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    -> LEFT JOIN Payments p       ON p.cycle_id = mc.cycle_id AND p.user_id = gm.user_id
    -> GROUP BY u.user_id, u.name, cg.group_id, cg.group_name;
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE OR REPLACE VIEW vw_cycle_results AS
   -> SELECT mc.cycle_id, cg.group_name, mc.month_number, mc.cycle_date, mc.status,
   ->        w.name AS winner, mc.winning_bid_amount, po.amount_disbursed, po.dividend_per_member
   -> FROM MonthlyCycles mc
   -> JOIN ChitGroups cg ON cg.group_id = mc.group_id
   -> LEFT JOIN Payouts po ON po.cycle_id = mc.cycle_id
   -> LEFT JOIN Users w    ON w.user_id  = po.winner_id;
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE OR REPLACE VIEW vw_defaulter_report AS
    -> SELECT d.default_id, u.name, u.phone, cg.group_name, mc.month_number,
    ->        d.amount_due, d.penalty_applied, d.resolved_status
    -> FROM Defaulters d
    -> JOIN Users u         ON u.user_id   = d.user_id
    -> JOIN ChitGroups cg   ON cg.group_id = d.group_id
    -> JOIN MonthlyCycles mc ON mc.cycle_id = d.cycle_id;
Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CREATE OR REPLACE VIEW vw_bid_ranking AS
    -> SELECT b.cycle_id, u.name, b.bid_amount, b.bid_time, b.is_winner,
    ->        RANK() OVER (PARTITION BY b.cycle_id ORDER BY b.bid_amount DESC) AS bid_rank
    -> FROM Bids b JOIN Users u ON u.user_id = b.user_id;
Query OK, 0 rows affected (0.02 sec)
```

# Running Business Workflow

```
PES2UG24CS460_PES2UG24CS435_G10$CALL sp_close_cycle(1);
+----------+-----------+-------------+------------------+--------------------+
| cycle_id | winner_id | winning_bid | amount_disbursed | dividend_per_member |
+----------+-----------+-------------+------------------+--------------------+
|        1 |         5 |     9500.00 |         90500.00 |             1187.50 |
+----------+-----------+-------------+------------------+--------------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.03 sec)

PES2UG24CS460_PES2UG24CS435_G10$CALL sp_close_cycle(2);
+----------+-----------+-------------+------------------+--------------------+
| cycle_id | winner_id | winning_bid | amount_disbursed | dividend_per_member |
+----------+-----------+-------------+------------------+--------------------+
|        2 |         7 |     9200.00 |         90800.00 |             1150.00 |
+----------+-----------+-------------+------------------+--------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

PES2UG24CS460_PES2UG24CS435_G10$CALL sp_close_cycle(4);
+----------+-----------+-------------+------------------+--------------------+
| cycle_id | winner_id | winning_bid | amount_disbursed | dividend_per_member |
+----------+-----------+-------------+------------------+--------------------+
|        4 |         5 |     3500.00 |         26500.00 |              583.33 |
+----------+-----------+-------------+------------------+--------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$UPDATE Payments SET status = 'defaulted', penalty_amount = 500
    -> WHERE cycle_id = 3 AND user_id = 10;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$UPDATE Defaulters d JOIN ChitGroups g ON g.group_id = d.group_id
    -> SET d.amount_due = g.monthly_installment WHERE d.amount_due = 0;   -- fix amount_due
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$CALL sp_record_payment(5, 7, 5000, '2026-03-09');
+--------------------------+
| result                   |
+--------------------------+
| Payment recorded as paid |
+--------------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CALL sp_mark_defaulters(5);
+--------------------------+
| members_marked_defaulter |
+--------------------------+
|                        2 |
+--------------------------+
1 row in set (0.05 sec)

Query OK, 0 rows affected (0.06 sec)

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$CALL sp_add_member(3, 4);
+--------------------------+
| result                   |
+--------------------------+
| User 4 added to group 3 |
+--------------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CALL sp_add_member(3, 5);
+--------------------------+
| result                   |
+--------------------------+
| User 5 added to group 3 |
+--------------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CALL sp_add_member(3, 6);
+--------------------------+
| result                   |
+--------------------------+
| User 6 added to group 3 |
+--------------------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$CALL sp_generate_cycles(3);
+---------------------------------+
| result                          |
+---------------------------------+
| 12 cycles generated for group 3 |
+---------------------------------+
1 row in set (0.10 sec)

Query OK, 0 rows affected (0.10 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT fn_member_count(1)           AS members_group1,
    ->        fn_calc_penalty(12500, 3)    AS penalty_3_days_late,
    ->        fn_total_paid(4, 1)          AS arjun_total_paid_group1,
    ->        fn_dividend(1, 9500)         AS dividend_for_bid_9500,
    ->        fn_risk_level(10)            AS mohan_risk,
    ->        fn_risk_level(4)             AS arjun_risk;
+---------------+---------------------+-------------------------+----------------------+------------+------------+
| members_group1 | penalty_3_days_late | arjun_total_paid_group1 | dividend_for_bid_9500 | mohan_risk | arjun_risk |
+---------------+---------------------+-------------------------+----------------------+------------+------------+
|              8 |              280.00 |                37500.00 |               1187.50 | MEDIUM     | LOW        |
+---------------+---------------------+-------------------------+----------------------+------------+------------+
1 row in set (0.00 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$CALL sp_group_summary(1);
+----------+--------------------+-------------+--------------+--------------------+-----------------+-------------+-----------------+-----------------+--------+
| group_id | group_name         | organizer   | total_amount | monthly_installment | duration_months | max_members | current_members | cycles_completed | status |
+----------+--------------------+-------------+--------------+--------------------+-----------------+-------------+-----------------+-----------------+--------+
|        1 | Gold Savers Circle | Ramesh Kumar |    100000.00 |           12500.00 |               8 |           8 |               8 |               2 | active |
+----------+--------------------+-------------+--------------+--------------------+-----------------+-------------+-----------------+-----------------+--------+
1 row in set (0.01 sec)

+---------------+-------------+----------------+
| name          | slot_status | has_won_payout |
+---------------+-------------+----------------+
| Lakshmi Devi  | active      |              0 |
| Arjun Nair    | active      |              0 |
| Priya Sharma  | active      |              1 |
| Karthik Reddy | active      |              0 |
| Sneha Iyer    | active      |              1 |
| Rahul Verma   | active      |              0 |
| Divya Menon   | active      |              0 |
| Mohan Das     | active      |              0 |
+---------------+-------------+----------------+
8 rows in set (0.02 sec)

+--------------+------------+---------+--------------------+
| month_number | cycle_date | status  | winning_bid_amount |
+--------------+------------+---------+--------------------+
|            1 | 2026-01-05 | closed  |            9500.00 |
|            2 | 2026-02-05 | closed  |            9200.00 |
|            3 | 2026-03-05 | bidding |               NULL |
+--------------+------------+---------+--------------------+
3 rows in set (0.05 sec)

Query OK, 0 rows affected (0.06 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO ChitGroups (group_id, group_name, organizer_id, total_amount, monthly_installment,
    ->                         duration_months, max_members, start_date)
    -> VALUES (4, 'Temporary Demo Group', 2, 50000, 10000, 5, 5, '2026-11-01');
Query OK, 1 row affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$
PES2UG24CS460_PES2UG24CS435_G10$UPDATE ChitGroups SET status = 'cancelled' WHERE group_id = 4;       -- trg_audit_group_status
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM AuditLog;
+--------+------------+-----------+---------------+------------+------------+---------------------+
| log_id | table_name | record_id | action        | old_value | new_value | changed_at          |
+--------+------------+-----------+---------------+------------+------------+---------------------+
|      1 | ChitGroups |         4 | STATUS_CHANGE | upcoming  | cancelled | 2026-10-04 14:35:17 |
+--------+------------+-----------+---------------+------------+------------+---------------------+
1 row in set (0.00 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$DELETE FROM ChitGroups WHERE group_id = 4;
Query OK, 1 row affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$DELETE FROM Notifications WHERE read_status = TRUE;
Query OK, 2 rows affected (0.02 sec)
```

# Select Queries

INNER JOIN

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT cg.group_name, u.name AS member, gm.joined_date, gm.slot_status, gm.has_won_payout
    -> FROM GroupMembers gm
    -> JOIN ChitGroups cg ON cg.group_id = gm.group_id
    -> JOIN Users u       ON u.user_id   = gm.user_id
    -> ORDER BY cg.group_name, u.name;
+----------------------+---------------+-------------+-------------+----------------+
| group_name           | member        | joined_date | slot_status | has_won_payout |
+----------------------+---------------+-------------+-------------+----------------+
| Gold Savers Circle   | Arjun Nair    | 2025-12-21  | active      |              0 |
| Gold Savers Circle   | Divya Menon   | 2025-12-25  | active      |              0 |
| Gold Savers Circle   | Karthik Reddy | 2025-12-22  | active      |              0 |
| Gold Savers Circle   | Lakshmi Devi  | 2025-12-20  | active      |              0 |
| Gold Savers Circle   | Mohan Das     | 2025-12-26  | active      |              0 |
| Gold Savers Circle   | Priya Sharma  | 2025-12-22  | active      |              1 |
| Gold Savers Circle   | Rahul Verma   | 2025-12-24  | active      |              0 |
| Gold Savers Circle   | Sneha Iyer    | 2025-12-23  | active      |              1 |
| Platinum Future Fund | Arjun Nair    | 2026-10-04  | active      |              0 |
| Platinum Future Fund | Karthik Reddy | 2026-10-04  | active      |              0 |
| Platinum Future Fund | Priya Sharma  | 2026-10-04  | active      |              0 |
| Silver Monthly Club  | Arjun Nair    | 2026-01-15  | active      |              0 |
| Silver Monthly Club  | Divya Menon   | 2026-01-20  | active      |              0 |
| Silver Monthly Club  | Karthik Reddy | 2026-01-17  | active      |              0 |
| Silver Monthly Club  | Priya Sharma  | 2026-01-16  | active      |              1 |
| Silver Monthly Club  | Rahul Verma   | 2026-01-19  | active      |              0 |
| Silver Monthly Club  | Sneha Iyer    | 2026-01-18  | active      |              0 |
+----------------------+---------------+-------------+-------------+----------------+
17 rows in set (0.00 sec)
```

LEFT JOIN

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.user_id, u.name, u.role, COUNT(gm.membership_id) AS groups_joined
    -> FROM Users u LEFT JOIN GroupMembers gm ON gm.user_id = u.user_id
    -> GROUP BY u.user_id, u.name, u.role
    -> ORDER BY groups_joined DESC, u.name;
+---------+---------------+-----------+---------------+
| user_id | name          | role      | groups_joined |
+---------+---------------+-----------+---------------+
|       4 | Arjun Nair    | member    |             3 |
|       6 | Karthik Reddy | member    |             3 |
|       5 | Priya Sharma  | member    |             3 |
|       9 | Divya Menon   | member    |             2 |
|       8 | Rahul Verma   | member    |             2 |
|       7 | Sneha Iyer    | member    |             2 |
|       3 | Lakshmi Devi  | organizer |             1 |
|      10 | Mohan Das     | member    |             1 |
|       1 | Admin Rao     | admin     |             0 |
|       2 | Ramesh Kumar  | organizer |             0 |
|      11 | Test User     | member    |             0 |
+---------+---------------+-----------+---------------+
11 rows in set (0.00 sec)
```

GROUP BY + AGGREAGTES

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT cg.group_name, COUNT(DISTINCT mc.cycle_id) AS cycles_with_payments,
    ->        COUNT(p.payment_id) AS payments, SUM(p.amount_paid) AS total_collected,
    ->        AVG(p.amount_paid) AS avg_payment, MAX(p.penalty_amount) AS max_penalty
    -> FROM ChitGroups cg
    -> JOIN MonthlyCycles mc ON mc.group_id = cg.group_id
    -> JOIN Payments p       ON p.cycle_id  = mc.cycle_id
    -> GROUP BY cg.group_id, cg.group_name;
+--------------------+---------------------+----------+-----------------+-------------+-------------+
| group_name         | cycles_with_payments | payments | total_collected | avg_payment | max_penalty |
+--------------------+---------------------+----------+-----------------+-------------+-------------+
| Gold Savers Circle |                   3 |       24 |       287500.00 | 11979.166667 |      500.00 |
| Silver Monthly Club |                   2 |       12 |        50000.00 |  4166.666667 |      100.00 |
+--------------------+---------------------+----------+-----------------+-------------+-------------+
2 rows in set (0.01 sec)
```

HAVING

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.name, COUNT(*) AS bad_payments, SUM(p.penalty_amount) AS total_penalty
    -> FROM Payments p JOIN Users u ON u.user_id = p.user_id
    -> WHERE p.status IN ('late', 'defaulted')
    -> GROUP BY u.user_id, u.name
    -> HAVING SUM(p.penalty_amount) > 0
    -> ORDER BY total_penalty DESC;
+-------------+--------------+---------------+
| name        | bad_payments | total_penalty |
+-------------+--------------+---------------+
| Mohan Das   |            1 |        500.00 |
| Rahul Verma |            1 |        100.00 |
| Divya Menon |            1 |        100.00 |
+-------------+--------------+---------------+
3 rows in set (0.00 sec)
```

ORDER BY + LIMIT

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT cg.group_name, mc.month_number, u.name, b.bid_amount
    -> FROM Bids b
    -> JOIN MonthlyCycles mc ON mc.cycle_id = b.cycle_id
    -> JOIN ChitGroups cg    ON cg.group_id = mc.group_id
    -> JOIN Users u          ON u.user_id   = b.user_id
    -> ORDER BY b.bid_amount DESC LIMIT 5;
+--------------------+--------------+---------------+------------+
| group_name         | month_number | name          | bid_amount |
+--------------------+--------------+---------------+------------+
| Gold Savers Circle |            1 | Priya Sharma  |    9500.00 |
| Gold Savers Circle |            2 | Sneha Iyer    |    9200.00 |
| Gold Savers Circle |            3 | Karthik Reddy |    9100.00 |
| Gold Savers Circle |            3 | Lakshmi Devi  |    9000.00 |
| Gold Savers Circle |            1 | Sneha Iyer    |    9000.00 |
+--------------------+--------------+---------------+------------+
5 rows in set (0.00 sec)
```

SCALAR SUBQUERY

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT cycle_id, user_id, bid_amount
    -> FROM Bids WHERE bid_amount > (SELECT AVG(bid_amount) FROM Bids)
    -> ORDER BY bid_amount DESC;
+----------+---------+------------+
| cycle_id | user_id | bid_amount |
+----------+---------+------------+
|        1 |       5 |    9500.00 |
|        2 |       7 |    9200.00 |
|        3 |       6 |    9100.00 |
|        1 |       7 |    9000.00 |
|        3 |       3 |    9000.00 |
|        3 |       4 |    8800.00 |
|        2 |       6 |    8500.00 |
|        1 |       4 |    8000.00 |
|        2 |       8 |    7500.00 |
+----------+---------+------------+
9 rows in set (0.00 sec)
```

IN SUBQUERY

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT user_id, name FROM Users
    -> WHERE user_id IN (SELECT winner_id FROM Payouts);
+---------+--------------+
| user_id | name         |
+---------+--------------+
|       5 | Priya Sharma |
|       7 | Sneha Iyer   |
+---------+--------------+
2 rows in set (0.00 sec)
```

NOT EXIST

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.user_id, u.name FROM Users u
    -> WHERE u.role = 'member'
    ->   AND NOT EXISTS (SELECT 1 FROM Defaulters d WHERE d.user_id = u.user_id);
+---------+---------------+
| user_id | name          |
+---------+---------------+
|       4 | Arjun Nair    |
|       5 | Priya Sharma  |
|       6 | Karthik Reddy |
|       7 | Sneha Iyer    |
|      11 | Test User     |
+---------+---------------+
5 rows in set (0.00 sec)
```

CORREALTED SUBQUERY

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT b.cycle_id, b.user_id, b.bid_amount
    -> FROM Bids b
    -> WHERE b.bid_amount = (SELECT MAX(b2.bid_amount) FROM Bids b2 WHERE b2.cycle_id = b.cycle_id)
    -> ORDER BY b.cycle_id;
+----------+---------+------------+
| cycle_id | user_id | bid_amount |
+----------+---------+------------+
|        1 |       5 |    9500.00 |
|        2 |       7 |    9200.00 |
|        3 |       6 |    9100.00 |
|        4 |       5 |    3500.00 |
+----------+---------+------------+
4 rows in set (0.00 sec)
```

CORREALTED EXISTS

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.name FROM Users u
    -> WHERE EXISTS (SELECT 1 FROM Payments p WHERE p.user_id = u.user_id AND p.status = 'late');
Empty set (0.00 sec)
```

CORREALTED+AGGREAGATES

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT p.payment_id, p.user_id, p.cycle_id, p.amount_paid
    -> FROM Payments p
    -> WHERE p.amount_paid > (SELECT AVG(p2.amount_paid) FROM Payments p2 WHERE p2.user_id = p.user_id);
+------------+---------+----------+-------------+
| payment_id | user_id | cycle_id | amount_paid |
+------------+---------+----------+-------------+
|          2 |       4 |        1 |    12500.00 |
|          3 |       5 |        1 |    12500.00 |
|          4 |       6 |        1 |    12500.00 |
|          5 |       7 |        1 |    12500.00 |
|          6 |       8 |        1 |    12500.00 |
|          7 |       9 |        1 |    12500.00 |
|          8 |      10 |        1 |    12500.00 |
|         17 |       4 |        2 |    12500.00 |
|         18 |       5 |        2 |    12500.00 |
|         19 |       6 |        2 |    12500.00 |
|         20 |       7 |        2 |    12500.00 |
|         21 |       8 |        2 |    12500.00 |
|         22 |       9 |        2 |    12500.00 |
|         23 |      10 |        2 |    12500.00 |
|         32 |       4 |        3 |    12500.00 |
|         33 |       5 |        3 |    12500.00 |
|         34 |       6 |        3 |    12500.00 |
|         35 |       7 |        3 |    12500.00 |
|         36 |       8 |        3 |    12500.00 |
|         37 |       9 |        3 |    12500.00 |
+------------+---------+----------+-------------+
20 rows in set (0.00 sec)
```

NESTED

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT g.group_id, g.total
    -> FROM (SELECT mc.group_id, SUM(p.amount_paid) AS total
    ->       FROM MonthlyCycles mc JOIN Payments p ON p.cycle_id = mc.cycle_id
    ->       GROUP BY mc.group_id) g
    -> WHERE g.total > (SELECT AVG(x.total)
    ->                  FROM (SELECT SUM(p.amount_paid) AS total
    ->                        FROM MonthlyCycles mc JOIN Payments p ON p.cycle_id = mc.cycle_id
    ->                        GROUP BY mc.group_id) x);
+----------+-----------+
| group_id | total     |
+----------+-----------+
|        1 | 287500.00 |
+----------+-----------+
1 row in set (0.00 sec)
```

ANY/ALL

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT user_id, bid_amount FROM Bids
    -> WHERE cycle_id = 3 AND bid_amount > ANY (SELECT bid_amount FROM Bids WHERE cycle_id = 1);
+---------+------------+
| user_id | bid_amount |
+---------+------------+
|       3 |    9000.00 |
|       4 |    8800.00 |
|       6 |    9100.00 |
+---------+------------+
3 rows in set (0.00 sec)
```

UNION

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.name, 'Organizer' AS involvement FROM Users u
    -> WHERE u.user_id IN (SELECT organizer_id FROM ChitGroups)
    -> UNION
    -> SELECT u.name, 'Winner' FROM Users u
    -> WHERE u.user_id IN (SELECT winner_id FROM Payouts);
+--------------+-------------+
| name         | involvement |
+--------------+-------------+
| Ramesh Kumar | Organizer   |
| Lakshmi Devi | Organizer   |
| Priya Sharma | Winner      |
| Sneha Iyer   | Winner      |
+--------------+-------------+
4 rows in set (0.00 sec)
```

CASE+FUCNTIONS

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT u.name, fn_risk_level(u.user_id) AS risk,
    ->        CASE WHEN fn_risk_level(u.user_id) = 'LOW'    THEN 'Eligible for all groups'
    ->             WHEN fn_risk_level(u.user_id) = 'MEDIUM' THEN 'Needs monitoring'
    ->             ELSE 'Restrict new memberships' END AS recommendation
    -> FROM Users u WHERE u.role = 'member';
+---------------+--------+------------------------+
| name          | risk   | recommendation         |
+---------------+--------+------------------------+
| Arjun Nair    | LOW    | Eligible for all groups |
| Priya Sharma  | LOW    | Eligible for all groups |
| Karthik Reddy | LOW    | Eligible for all groups |
| Sneha Iyer    | LOW    | Eligible for all groups |
| Rahul Verma   | MEDIUM | Needs monitoring        |
| Divya Menon   | MEDIUM | Needs monitoring        |
| Mohan Das     | MEDIUM | Needs monitoring        |
| Test User     | LOW    | Eligible for all groups |
+---------------+--------+------------------------+
8 rows in set (0.00 sec)
```

MULTI TABLE JOIN

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT cg.group_name, mc.month_number, w.name AS winner, po.amount_disbursed,
    ->        po.dividend_per_member, mc.total_pool_amount
    -> FROM Payouts po
    -> JOIN MonthlyCycles mc ON mc.cycle_id = po.cycle_id
    -> JOIN ChitGroups cg    ON cg.group_id = mc.group_id
    -> JOIN Users w          ON w.user_id   = po.winner_id
    -> ORDER BY cg.group_id, mc.month_number;
+--------------------+--------------+--------------+------------------+--------------------+-------------------+
| group_name         | month_number | winner       | amount_disbursed | dividend_per_member | total_pool_amount |
+--------------------+--------------+--------------+------------------+--------------------+-------------------+
| Gold Savers Circle |            1 | Priya Sharma |         90500.00 |            1187.50 |         100000.00 |
| Gold Savers Circle |            2 | Sneha Iyer   |         90800.00 |            1150.00 |         100000.00 |
| Silver Monthly Club |           1 | Priya Sharma |         26500.00 |             583.33 |          30000.00 |
+--------------------+--------------+--------------+------------------+--------------------+-------------------+
3 rows in set (0.00 sec)
```

FULL TEXT SEARCH

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT notif_id, user_id, message
    -> FROM Notifications WHERE MATCH(message) AGAINST('+payment +reminder' IN BOOLEAN MODE);
+----------+---------+-----------------------------------------------------+
| notif_id | user_id | message                                             |
+----------+---------+-----------------------------------------------------+
|        3 |       6 | Payment reminder: monthly installment is due.       |
|        5 |       8 | Payment reminder: please pay before the cycle date. |
+----------+---------+-----------------------------------------------------+
2 rows in set (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT notif_id, user_id, message
    -> FROM Notifications WHERE MATCH(message) AGAINST('payout congratulations' IN BOOLEAN MODE);
+----------+---------+-------------------------------------------------------------------------------------+
| notif_id | user_id | message                                                                             |
+----------+---------+-------------------------------------------------------------------------------------+
|        6 |       5 | Congratulations! You won the auction for cycle 1. Payout amount: Rs. 90500.00 |
|        7 |       7 | Congratulations! You won the auction for cycle 2. Payout amount: Rs. 90800.00 |
|        8 |       5 | Congratulations! You won the auction for cycle 4. Payout amount: Rs. 26500.00 |
+----------+---------+-------------------------------------------------------------------------------------+
3 rows in set (0.00 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM vw_group_overview;
+----------+--------------------+-----------+--------------+---------------------+-----------------+-------------+-----------------+-----------------+----------+
| group_id | group_name         | organizer | total_amount | monthly_installment | duration_months | max_members | current_members | cycles_completed | status   |
+----------+--------------------+-----------+--------------+---------------------+-----------------+-------------+-----------------+-----------------+----------+
|        1 | Gold Savers Circle | Ramesh Kumar |    100000.00 |            12500.00 |               8 |           8 |               8 |               2 | active   |
|        2 | Silver Monthly Club | Lakshmi Devi |     30000.00 |             5000.00 |               6 |           6 |               6 |               1 | active   |
|        3 | Platinum Future Fund | Ramesh Kumar |    240000.00 |            20000.00 |              12 |          12 |               3 |               0 | upcoming |
+----------+--------------------+-----------+--------------+---------------------+-----------------+-------------+-----------------+-----------------+----------+
3 rows in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM vw_member_payment_summary ORDER BY total_penalty DESC;
+---------+-------------+----------+--------------------+----------------+-------------+---------------+-------------+---------------+
| user_id | name        | group_id | group_name         | payments_made  | total_paid  | total_penalty | late_count  | default_count |
+---------+-------------+----------+--------------------+----------------+-------------+---------------+-------------+---------------+
|      10 | Mohan Das   |        1 | Gold Savers Circle |              3 |    25000.00 |        500.00 |           0 |             1 |
|       9 | Divya Menon |        2 | Silver Monthly Club |              2 |     5000.00 |        100.00 |           0 |             1 |
|       8 | Rahul Verma |        2 | Silver Monthly Club |              2 |     5000.00 |        100.00 |           0 |             1 |
|       9 | Divya Menon |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       8 | Rahul Verma |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       7 | Sneha Iyer  |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       6 | Karthik Reddy |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       5 | Priya Sharma |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       4 | Arjun Nair  |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       3 | Lakshmi Devi |        1 | Gold Savers Circle |              3 |    37500.00 |          0.00 |           0 |             0 |
|       7 | Sneha Iyer  |        2 | Silver Monthly Club |              2 |    10000.00 |          0.00 |           0 |             0 |
|       6 | Karthik Reddy |        2 | Silver Monthly Club |              2 |    10000.00 |          0.00 |           0 |             0 |
|       5 | Priya Sharma |        2 | Silver Monthly Club |              2 |    10000.00 |          0.00 |           0 |             0 |
|       4 | Arjun Nair  |        2 | Silver Monthly Club |              2 |    10000.00 |          0.00 |           0 |             0 |
|       6 | Karthik Reddy |        3 | Platinum Future Fund |              0 |        0.00 |          0.00 |           0 |             0 |
|       5 | Priya Sharma |        3 | Platinum Future Fund |              0 |        0.00 |          0.00 |           0 |             0 |
|       4 | Arjun Nair  |        3 | Platinum Future Fund |              0 |        0.00 |          0.00 |           0 |             0 |
+---------+-------------+----------+--------------------+----------------+-------------+---------------+-------------+---------------+
17 rows in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM vw_cycle_results;
+----------+--------------------+--------------+------------+------------+-------------+--------------------+-----------------+---------------------+
| cycle_id | group_name         | month_number | cycle_date | status     | winner      | winning_bid_amount | amount_disbursed | dividend_per_member |
+----------+--------------------+--------------+------------+------------+-------------+--------------------+-----------------+---------------------+
|        1 | Gold Savers Circle |            1 | 2026-01-05 | closed     | Priya Sharma |            9500.00 |         90500.00 |             1187.50 |
|        2 | Gold Savers Circle |            2 | 2026-02-05 | closed     | Sneha Iyer  |            9200.00 |         90800.00 |             1150.00 |
|        3 | Gold Savers Circle |            3 | 2026-03-05 | bidding    | NULL        |               NULL |             NULL |                NULL |
|        4 | Silver Monthly Club |            1 | 2026-02-10 | closed     | Priya Sharma |            3500.00 |         26500.00 |              583.33 |
|        5 | Silver Monthly Club |            2 | 2026-03-10 | collection | NULL        |               NULL |             NULL |                NULL |
|        6 | Platinum Future Fund |            1 | 2026-12-01 | collection | NULL        |               NULL |             NULL |                NULL |
|        7 | Platinum Future Fund |            2 | 2027-01-01 | collection | NULL        |               NULL |             NULL |                NULL |
|        8 | Platinum Future Fund |            3 | 2027-02-01 | collection | NULL        |               NULL |             NULL |                NULL |
|        9 | Platinum Future Fund |            4 | 2027-03-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       10 | Platinum Future Fund |            5 | 2027-04-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       11 | Platinum Future Fund |            6 | 2027-05-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       12 | Platinum Future Fund |            7 | 2027-06-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       13 | Platinum Future Fund |            8 | 2027-07-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       14 | Platinum Future Fund |            9 | 2027-08-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       15 | Platinum Future Fund |           10 | 2027-09-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       16 | Platinum Future Fund |           11 | 2027-10-01 | collection | NULL        |               NULL |             NULL |                NULL |
|       17 | Platinum Future Fund |           12 | 2027-11-01 | collection | NULL        |               NULL |             NULL |                NULL |
+----------+--------------------+--------------+------------+------------+-------------+--------------------+-----------------+---------------------+
17 rows in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM vw_defaulter_report;
+------------+-----------+------------+--------------------+--------------+------------+-----------------+-----------------+
| default_id | name      | phone      | group_name         | month_number | amount_due | penalty_applied | resolved_status |
+------------+-----------+------------+--------------------+--------------+------------+-----------------+-----------------+
|          1 | Mohan Das | 9000000010 | Gold Savers Circle |            3 |   12500.00 |          500.00 | pending         |
|          2 | Rahul Verma | 9000000008 | Silver Monthly Club |            2 |    5000.00 |          100.00 | pending         |
|          3 | Divya Menon | 9000000009 | Silver Monthly Club |            2 |    5000.00 |          100.00 | pending         |
+------------+-----------+------------+--------------------+--------------+------------+-----------------+-----------------+
3 rows in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM vw_bid_ranking WHERE bid_rank <= 2;
+----------+-------------+------------+---------------------+------------+----------+
| cycle_id | name        | bid_amount | bid_time            | is_winner  | bid_rank |
+----------+-------------+------------+---------------------+------------+----------+
|        1 | Priya Sharma |    9500.00 | 2026-01-05 10:05:00 |          1 |        1 |
|        1 | Sneha Iyer  |    9000.00 | 2026-01-05 10:15:00 |          0 |        2 |
|        2 | Sneha Iyer  |    9200.00 | 2026-02-05 10:07:00 |          1 |        1 |
|        2 | Karthik Reddy |    8500.00 | 2026-02-05 10:00:00 |          0 |        2 |
|        3 | Karthik Reddy |    9100.00 | 2026-03-05 10:15:00 |          0 |        1 |
|        3 | Lakshmi Devi |    9000.00 | 2026-03-05 10:00:00 |          0 |        2 |
|        4 | Priya Sharma |    3500.00 | 2026-02-10 11:05:00 |          1 |        1 |
|        4 | Arjun Nair  |    3000.00 | 2026-02-10 11:00:00 |          0 |        2 |
+----------+-------------+------------+---------------------+------------+----------+
8 rows in set (0.00 sec)
```

```
PES2UG24CS460_PES2UG24CS435_G10$EXPLAIN SELECT * FROM Bids WHERE cycle_id = 3 ORDER BY bid_amount DESC;
+----+-------------+-------+------------+------+-------------------------------+-------------------------+---------+-------+------+----------+-------+
| id | select_type | table | partitions | type | possible_keys                 | key                     | key_len | ref   | rows | filtered | Extra |
+----+-------------+-------+------------+------+-------------------------------+-------------------------+---------+-------+------+----------+-------+
|  1 | SIMPLE      | Bids  | NULL       | ref  | cycle_id,idx_bids_cycle_amount | idx_bids_cycle_amount | 4       | const |    3 |   100.00 | NULL  |
+----+-------------+-------+------------+------+-------------------------------+-------------------------+---------+-------+------+----------+-------+
1 row in set, 1 warning (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$EXPLAIN SELECT * FROM Payments WHERE status = 'late' AND cycle_id = 2;
+----+-------------+----------+------------+------+-------------------------------+-------------------------+---------+-------------+------+----------+-----------------------+
| id | select_type | table    | partitions | type | possible_keys                 | key                     | key_len | ref         | rows | filtered | Extra                 |
+----+-------------+----------+------------+------+-------------------------------+-------------------------+---------+-------------+------+----------+-----------------------+
|  1 | SIMPLE      | Payments | NULL       | ref  | cycle_id,idx_payments_status_cycle | idx_payments_status_cycle | 6       | const,const |    1 |   100.00 | Using index condition |
+----+-------------+----------+------------+------+-------------------------------+-------------------------+---------+-------------+------+----------+-----------------------+
1 row in set, 1 warning (0.00 sec)
```

# Transaction

COMMIT

```
PES2UG24CS460_PES2UG24CS435_G10$START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$UPDATE Payments SET amount_paid = 12500, status = 'late', payment_date = '2026-03-15'
    -> WHERE cycle_id = 3 AND user_id = 10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$UPDATE Defaulters SET resolved_status = 'resolved' WHERE cycle_id = 3 AND user_id = 10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Notifications (user_id, message, type)
    -> VALUES (10, 'Your overdue installment for cycle 3 has been settled. Thank you.', 'payment');
Query OK, 1 row affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$COMMIT;
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM Defaulters WHERE user_id = 10;
+------------+---------+----------+----------+------------+-----------------+-----------------+
| default_id | user_id | group_id | cycle_id | amount_due | penalty_applied | resolved_status |
+------------+---------+----------+----------+------------+-----------------+-----------------+
|          1 |      10 |        1 |        3 |   12500.00 |          500.00 | resolved        |
+------------+---------+----------+----------+------------+-----------------+-----------------+
1 row in set (0.00 sec)
```

ROLLBACK

```
PES2UG24CS460_PES2UG24CS435_G10$START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$UPDATE Payments SET amount_paid = 0 WHERE cycle_id = 1;
Query OK, 8 rows affected (0.00 sec)
Rows matched: 8  Changed: 8  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$SELECT SUM(amount_paid) AS pool_inside_txn FROM Payments WHERE cycle_id = 1;
+-----------------+
| pool_inside_txn |
+-----------------+
|            0.00 |
+-----------------+
1 row in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT SUM(amount_paid) AS pool_after_rollback FROM Payments WHERE cycle_id = 1;
+---------------------+
| pool_after_rollback |
+---------------------+
|           100000.00 |
+---------------------+
1 row in set (0.00 sec)
```

SAVEPOINT

```
PES2UG24CS460_PES2UG24CS435_G10$START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Notifications (user_id, message, type) VALUES (4, 'Notice A: keep this message', 'general');
Query OK, 1 row affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SAVEPOINT after_notice_a;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$INSERT INTO Notifications (user_id, message, type) VALUES (4, 'Notice B: discard this message', 'general');
Query OK, 1 row affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$ROLLBACK TO SAVEPOINT after_notice_a;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$COMMIT;
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT message FROM Notifications WHERE user_id = 4;     -- only Notice A remains
+----------------------------+
| message                    |
+----------------------------+
| Notice A: keep this message |
+----------------------------+
1 row in set (0.00 sec)
```

ROW LOCKING + ISOLATION LEVEL

```
PES2UG24CS460_PES2UG24CS435_G10$SET TRANSACTION ISOLATION LEVEL REPEATABLE READ;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$SELECT * FROM MonthlyCycles WHERE cycle_id = 3 FOR UPDATE;   -- locks the row
+----------+----------+--------------+------------+---------+------------------+--------------------+
| cycle_id | group_id | month_number | cycle_date | status  | total_pool_amount | winning_bid_amount |
+----------+----------+--------------+------------+---------+------------------+--------------------+
|        3 |        1 |            3 | 2026-03-05 | bidding |             NULL |               NULL |
+----------+----------+--------------+------------+---------+------------------+--------------------+
1 row in set (0.00 sec)

PES2UG24CS460_PES2UG24CS435_G10$UPDATE MonthlyCycles SET cycle_date = '2026-03-06' WHERE cycle_id = 3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS460_PES2UG24CS435_G10$COMMIT;
Query OK, 0 rows affected (0.01 sec)
```